MAE - 5730, Nice Solution for Problem 23 (Approved by Prof. Ruina)

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1 Finding Periodic Orbits for 3 point masses moving under Inverse Square Gravity ($m_1 = m_2 = m_3 = 1$, $G = 1$)

1.1 Deriving the sets of differential equations for the system-

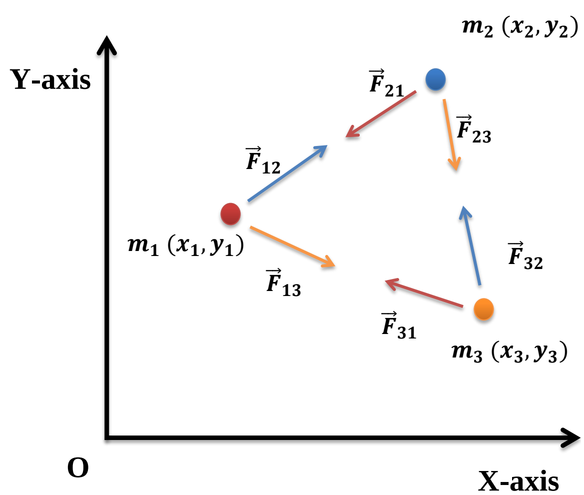

Figure 1: Free body diagram of m_1, m_2 and m_3

Let $\vec{\boldsymbol{F}}_{\boldsymbol{ij}}$ denote the force acting on mass-i due to mass-j.

Similarly, let $\vec{\boldsymbol{r}}_{\boldsymbol{ij}}$ denote the position vector from m_i to m_j.

$$\vec{\boldsymbol{r}}_{\boldsymbol{ij}} = (x_j - x_i)\hat{i} + (y_j - y_i)\hat{j} + (y_j - y_i)\hat{k} \ \ \forall\ i, j\ \in \{1, 2, 3\} \tag{1}$$

From inverse square law of gravity:

$$\vec{\boldsymbol{F}}_{\boldsymbol{ij}} = -\frac{Gm_i m_j}{|\vec{r}_{ij}|^3}\vec{r}_{ij} \ \ \forall\ i, j\ \in \{1, 2, 3\}\ such\ that\ i \neq\ j \tag{2}$$

From linear momentum balance for m_1, we get:

$$m_1\vec{\boldsymbol{a}}_{\boldsymbol{1}} = \vec{\boldsymbol{F}}_{1,net}$$

$$\Rightarrow m_1(\ddot{x}_1\hat{i} + \ddot{y}_1\hat{j} + \ddot{z}_1\hat{k}) = \vec{\boldsymbol{F}}_{\boldsymbol{12}} + \vec{\boldsymbol{F}}_{\boldsymbol{13}}$$

$$\Rightarrow \ddot{x}_1\hat{i} + \ddot{y}_1\hat{j} + \ddot{z}_1\hat{k} = \frac{Gm_2}{|\vec{r}_{12}|^3}\vec{r}_{12} + \frac{Gm_3}{|\vec{r}_{13}|^3}\vec{r}_{13}$$

$$\Rightarrow \ddot{x}_1\hat{i} + \ddot{y}_1\hat{j} + \ddot{z}_1\hat{k} = G\Big[\Big(\frac{m_2}{r_{12}^3}(x_2 - x_1) + \frac{m_3}{r_{13}^3}(x_3 - x_1)\Big)\hat{i} + \Big(\frac{m_2}{r_{12}^3}(y_2 - y_1) + \frac{m_3}{r_{13}^3}(y_3 - y_1)\Big)\hat{j} + \Big(\frac{m_2}{r_{12}^3}(z_2 - z_1) + \frac{m_3}{r_{13}^3}(z_3 - z_1)\Big)\hat{k}\Big] \tag{3}$$

From equation 3 we get three 2^{nd} order differential equations:

$$\ddot{x}_1 = G\Big(\frac{m_2}{r_{12}^3}(x_2 - x_1) + \frac{m_3}{r_{13}^3}(x_3 - x_1)\Big) \tag{4}$$

$$\ddot{y}_1 = G\Big(\frac{m_2}{r_{12}^3}(y_2 - y_1) + \frac{m_3}{r_{13}^3}(y_3 - y_1)\Big) \tag{5}$$

$$\ddot{z}_1 = G\Big(\frac{m_2}{r_{12}^3}(z_2 - z_1) + \frac{m_3}{r_{13}^3}(z_3 - z_1)\Big) \tag{6}$$

Similarly, by linear momentum balance for m_2, we get three 2^{nd} order differential equations:

$$\ddot{x}_2 = G\Big(\frac{m_1}{r_{12}^3}(x_1 - x_2) + \frac{m_3}{r_{23}^3}(x_3 - x_2)\Big) \tag{7}$$

$$\ddot{y}_2 = G\Big(\frac{m_1}{r_{12}^3}(y_1 - y_2) + \frac{m_3}{r_{23}^3}(y_3 - y_2)\Big) \tag{8}$$

$$\ddot{z}_2 = G\Big(\frac{m_1}{r_{12}^3}(z_1 - z_2) + \frac{m_3}{r_{23}^3}(z_3 - z_2)\Big) \tag{9}$$

Similarly, by linear momentum balance for m_2, we get three 2^{nd} order differential equations:

$$\ddot{x}_3 = G\Big(\frac{m_2}{r_{32}^3}(x_2 - x_3) + \frac{m_1}{r_{13}^3}(x_1 - x_3)\Big) \tag{10}$$

$$\ddot{y}_3 = G\Big(\frac{m_2}{r_{32}^3}(y_2 - y_3) + \frac{m_1}{r_{13}^3}(y_1 - y_3)\Big) \tag{11}$$

$$\ddot{z}_3 = G\Big(\frac{m_2}{r_{32}^3}(z_2 - z_3) + \frac{m_1}{r_{13}^3}(z_1 - z_3)\Big) \tag{12}$$

Equations 4 - 12 represent the set of differential equations for the given system dynamics. If we are given a set of initial conditions, we can generate the trajectories of the three masses using these set of differential equations.

1.2 Proposed Method to Find periodic Orbits by FMINCON

In order to find periodic orbits, I have posed the problem as a constrained minimization problem and solved it using MATLAB's inbuilt function FMINCON.

Let $[\boldsymbol{p}]_{12\times1}$ represent the state of the system.

$$\boldsymbol{p} = [x_1, y_1, x_2, y_2, x_3, y_3, \dot{x_1}, \dot{y_1}, \dot{x_2}, \dot{y_2}, \dot{x_3}, \dot{y_3}]^T$$

$$\Rightarrow \dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{p})$$

The variables for our minimization problem are the initial conditions for the states ($\boldsymbol{p}_o$) and time period (T). For a periodic orbit, the final states, after one time period, should converge to the initial states. So, $\|\boldsymbol{p}(T) - \boldsymbol{p}_o\| \rightarrow 0$. In addition, we can use bounds for the variables in order to achieve faster convergence of the minimization algorithm. So the final minimization problem is given by equation 13.

$$\begin{aligned} \min_{(\boldsymbol{p}_o, T)} f(\boldsymbol{p}_o, T) \; &= \; \|\boldsymbol{p}(T) - \boldsymbol{p}_o\| \\ \text{s.t.} \quad & Lb \leq \boldsymbol{p}_o \leq Ub \\ & T_{lower} \leq T \leq T_{upper} \end{aligned} \tag{13}$$

This optimization problem has multiple local minima. So one can understand that the solution obtained by $fmincon$ will heavily depend upon the guess chosen for the variables.

1.3 Solution-1

For the first set of guess for the optimization algorithm, as given in the problem itself, I obtained a periodic orbit as shown in figure 2.

$$\boldsymbol{p}_o = [-0.755,\ 0.355,\ 1.155,\ -0.0755,\ -0.4055,\ -0.3055,\ 0.9955,\ 0.07855,\ 0.1055,\ 0.4755,\ -1.1055,\ -0.5355]^T$$

$$T = 8$$

The bounds for search are:

$$\boldsymbol{p}_o \ \leq \ Ub = [-0.7,\ 0.4,\ 1.2,\ 0,\ 0.1,\ 0.1,\ 1,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 0.9]^T$$

$$\boldsymbol{p}_o \ \geq \ LB = [-0.9,\ 0.2,\ 0.9,\ -0.3,\ -0.5,\ -0.4,\ -0.5,\ -0.5,\ -0.5, -0.5,\ -1.2,\ -0.6]^T$$

$$6 \ \leq \ T \ \leq \ 8.2$$

So now equation 13 is complete and $fmincon$ can be used to minimize the problem. For implementation, I used the following parameters for solving equation 13 :

```
Algorithm = 'active-set';
MaxIter = 1500;
MaxFunEvals  = 2000;
TolX = 1e-7; TolCon = 1e-7; TolFun = 1e-7;
```

The solver converged after 47 iterations because the predicted change in the value of objective function in the next iteration, $f(\boldsymbol{p}_o, T)$, was less than $TolFun$. The values of initial conditions ($\boldsymbol{p}_o$) and time period (T) which will lead to periodic orbit, as obtained from the solution of the minimization problem are as follows:

$$x_1 = -0.7033,\ y_1 = 0.3949$$

$$x_2 = 1.1547,\ y_2 = -0.0932$$

$$x_3 = -0.3502,\ y_3 = -0.2804$$

$$V_{x1} = 0.9403,\ V_{y1} = 0.0714$$

$$V_{x2} = 0.1271,\ V_{y2} = 0.4545$$

$$V_{x3} = -1.0674,\ V_{y3} = -0.5260$$

$$T = 6.7880,\ \|\boldsymbol{p}(T) - \boldsymbol{p}_o\| = 0.00026$$

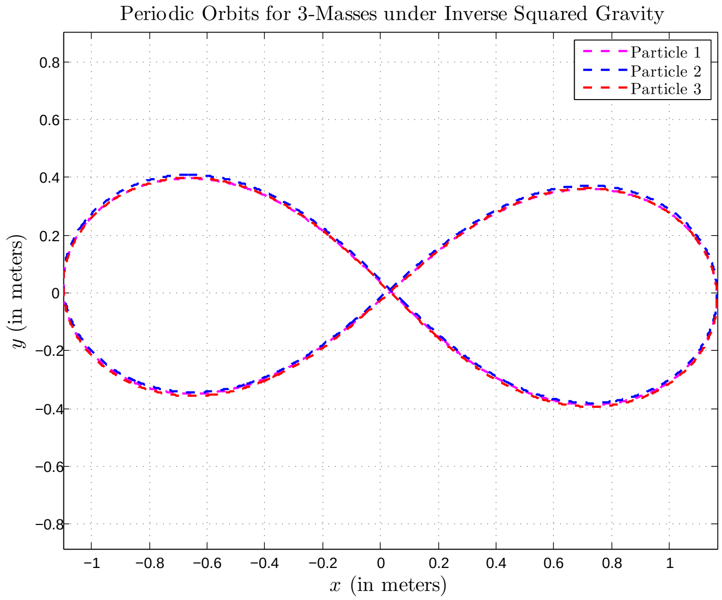

Figure 2: Periodic Orbit Obtained from Initial guess provided in section 1.3

1.4 Solution-2

Now I changed the guess for the initial conditions and time period, and obtained another converged result for periodic orbit which is shown in figure 3

$$\boldsymbol{p}_o \;=\; [-0.32,\ 0.42,\ -2,\ -0.45,\ 1,\ 0.42,\ 0,\ -1,\ 0,\ 0.1,\ 0,\ 0.8]^T$$
$$T = 11$$

The bounds for search are:

$$\boldsymbol{p}_o \;\leq\; Ub = [5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5]^T$$
$$\boldsymbol{p}_o \;\geq\; LB = [-5,\ -5,\ -5,\ -5,\ -5,\ -5,\ -5,\ -5,\ -5,\ -5,\ -5\ -5]^T$$
$$2 \;\leq\; T \;\leq\; 15$$

Again, I used the same optimization parameters for the solver as in section 1.3. In this case, the solver converged after 62 iterations because the predicted change in the value of objective function in the next iteration, $f(\boldsymbol{p}_o, T)$, was less than $TolFun$. The values of initial conditions ($\boldsymbol{p_o}$) and time period (T) which will lead to periodic orbit, as obtained from the solution of the minimization problem are as follows:

$$x_1 = -0.0828,\ y_1 = 0.3454$$
$$x_2 = -1.7272,\ y_2 = -0.1126$$
$$x_3 = 1.3181,\ y_3 = 0.1216$$
$$V_{x1} = 0.0892,\ V_{y1} = -0.9893$$
$$V_{x2} = -0.1160,\ V_{y2} = 0.1422$$
$$V_{x3} = 0.0268,\ V_{y3} = 0.8471$$
$$T = 11.3074,\ \|\boldsymbol{p}(T) - \boldsymbol{p}_o\| = 0.0005$$

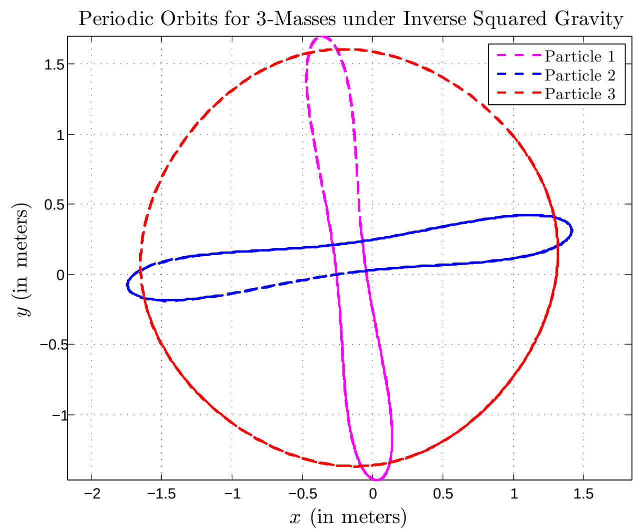

Figure 3: Periodic Orbit Obtained from Initial guess provided in section 1.4

1.5 Result Obtained from poor choice of tolerance

The three masses moving under inverse squared gravity results in periodic orbits only for some special initial conditions and if we choose any other initial condition, then they show a chaotic behaviour. For the optimization problem posed in section 1.2, if we choose a poor tolerance for FMINCON, then the solver doesn't succeed in obtaining the initial conditions corresponding to periodic orbits. In this section I have presented the solution obtained by choosing the following tolerance level for FMINCON:

```
TolX = 0.1; TolCon = 0.1; TolFun = 0.1;
```

Choosing such terrible tolerance level resulted in weird looking trajectories, as shown in figure 4 which are surely not periodic. The next section presents the MATLAB code that I developed in order for solving the problem.

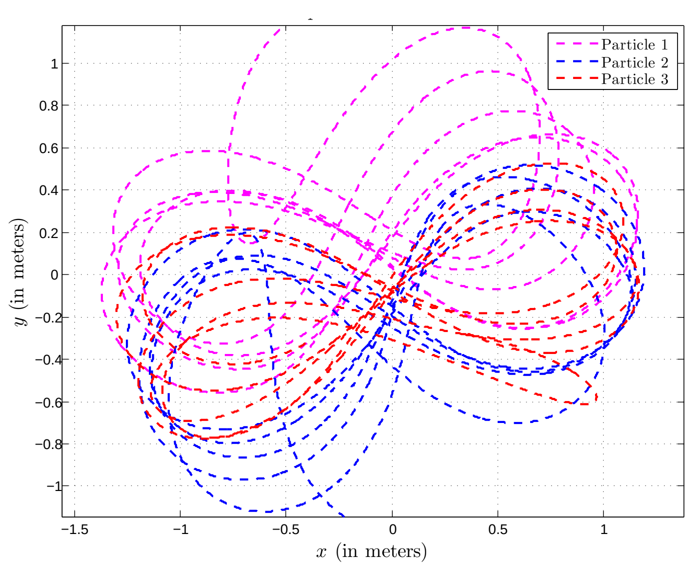

Figure 4: Trajectories obtained for poor tolerance, as chosen in section 1.5

1.6 MATLAB Code

```
%% Code written by Vikram Shree, Mechanical & Aerospace Engineering
%% Cornell University
%% MAE 5730, Nice Solution for Problem 23

%% This is the main file which needs to be run
% In this problem I have obtained periodic solutions for Three particles
% moving under Inverse Square Gravity by posing it as a optimization problem
% (Refer to text section-1.2)

clc;
close all;
clear all;

global G m1 m2 m3
% Setting the stucture containing the system parameters
m1 = 1; % in kg
m2 = 1; % in kg
m3 = 1; % in kg
G = 1;

%% Formulating the minimization problem as in section - 1.2 of text

% No Inequality Constraints,
A =[];
b = [];

% No Equality Constraints
Aeq = [];
beq = [];
% No non-linear constraints
nonlcon = [];

% Bounds on the variables
ub = [-0.7, 0.4, 1.2, 0, 0.1, 0.1, 1, 0.1, 0.2, 0.5, 1, 0.9, 8.2]';
lb = [-1, 0.2, 0.9, -0.3, -0.5, -0.4, -0.5, -0.5, -0.5, -0.5, -1.2, -0.6,
6]';

%% Setting Optimization Parameters

options = optimoptions('fmincon','Display','iter');
options.Algorithm = 'active-set';
options.MaxIter = 1500; % Max number of iterations
options.MaxFunEvals  = 2e3; % Max number of function evaluations

% Setting Solver Tolerance
options.TolX = 1e-7;
options.TolCon = 1e-7;
options.TolFun = 1e-7;

%%  This set of guesses for the solver
x1 = -0.755; y1 = 0.355;
x2 = 1.155; y2 = -0.0755;
x3 = -0.4055; y3 = -0.3055;
```

```
Vx1 = 0.9955; Vy1 = 0.07855;
Vx2 = 0.1055; Vy2 = 0.4755;
Vx3 = -1.1055; Vy3 = -0.5355;
T = 8; % Initial guess for Time period

x0 = [x1, y1, x2, y2, x3, y3, Vx1, Vy1, Vx2, Vy2, Vx3, Vy3, T];

% Solving the minimization problem
[x,fval] = fmincon(@f,x0,A,b,Aeq,beq,lb,ub,nonlcon,options)

%% Obtaining the trajectories for the 3 masses for converged initial
conditions.
% After obtaining the solution for minimization problem, I am using the
% converged result as initial condition for my system and obtined the
% trajectories to verify that they are indeed periodic.

inits = x(1:12); % Initial condition for ODE45
tf = 5*x(end); % I am interested in obtaining the trajectory for 5x(Time
Period)

% Setting Tolerances for ODE45
opts.RelTol = 1e-7;
opts.AbsTol = 1e-7;

% Obtaining numeriacal solution
[time, yArray] = ode45(@RHSfun,[0,tf],inits,opts);

% Extracting the positions of the 3-masses
trajX1 = yArray(:,1);
trajY1 = yArray(:,2);
trajX2 = yArray(:,3);
trajY2 = yArray(:,4);
trajX3 = yArray(:,5);
trajY3 = yArray(:,6);

%% Interpolating the result on a uniform time scale
delT = 0.1;
uniformTime = [0:delT:tf];

traj.X1 = interp1(time(:,1),yArray(:,1),uniformTime(:));
traj.Y1 = interp1(time(:,1),yArray(:,2),uniformTime(:));
traj.X2 = interp1(time(:,1),yArray(:,3),uniformTime(:));
traj.Y2 = interp1(time(:,1),yArray(:,4),uniformTime(:));
traj.X3 = interp1(time(:,1),yArray(:,5),uniformTime(:));
traj.Y3 = interp1(time(:,1),yArray(:,6),uniformTime(:));

%% Animation for periodic solution
delT = 0.1; % This is the time interval for the simulation
% setting it to '0.1' will show the positions of the masses at interval of
% about 0.1 seconds.
animation(traj, delT);
```

```
%% This function denotes the cost function which needs to be minimized
% This corresponds to 'f(po,T)' of equation - 13 in the text.

function res = errorNorm(x)

inits = x(1:12);
tf = x(13);
% Setting Tolerances for ODE45
opts.RelTol = 1e-6;
opts.AbsTol = 1e-6;

% Numeriacal solution
[time, yArray] = ode45(@RHSfun,[0,tf],inits,opts);

% Extracting the positions of the masses
xEnd = yArray(end,:);

% Returning the norm of the error which should be '0' for perfectly periodic
orbit
res = norm(inits-xEnd);
```

```
%% This function contains the system dynamics as in section - 1.1 of text

function yDot = RHSfun(t,y,para)
% Instead of supplying the parameters to 'RHSfun', it can directly access
% the parameters as I have declared them as 'Global' variables.

global G
global m1
global m2
global m3

r12 = sqrt((y(1)-y(3))^2 + (y(2)-y(4))^2);
r23 = sqrt((y(3)-y(5))^2 + (y(4)-y(6))^2);
r31 = sqrt((y(1)-y(5))^2 + (y(2)-y(6))^2);
r21 = r12;
r32 = r23;
r13 = r31;

% Writing the differential equation as in equation: (4)-(12), in text
% yDot = ddt([x1, y1, x2, y2, x3, y3, Vx1, Vy1, Vx2, Vy2, Vx3, Vy3]')

yDot(1,1) = y(7);
yDot(2,1) = y(8);
yDot(3,1) = y(9);
yDot(4,1) = y(10);
yDot(5,1) = y(11);
yDot(6,1) = y(12);
yDot(7,1) = -G*m2*(y(1)-y(3))/(r12^3) - G*m3*(y(1)-y(5))/(r13^3);
yDot(8,1) = -G*m2*(y(2)-y(4))/(r12^3) - G*m3*(y(2)-y(6))/(r13^3);
yDot(9,1) = -G*m1*(y(3)-y(1))/(r12^3) - G*m3*(y(3)-y(5))/(r23^3);
```

```
yDot(10,1) = -G*m1*(y(4)-y(2))/(r12^3) - G*m3*(y(4)-y(6))/(r23^3);
yDot(11,1) = -G*m1*(y(5)-y(1))/(r13^3) - G*m2*(y(5)-y(3))/(r23^3);
yDot(12,1) = -G*m1*(y(6)-y(2))/(r13^3) - G*m2*(y(6)-y(4))/(r23^3);
```

```
%% This function animates the trajectory of the three masses in periodic
orbits
% Input is the trajectory of the masses and the time interval for the
% simulation

function animation(traj,delT)

% 'traj' stores the trajectory of the masses
% 'delT' is the time step for simulation

fig = figure;
figure(fig)
n = size(traj.X1,1);
h1Old = plot(traj.X1(1),traj.Y1(2),'ob','LineWidth',1.5);
hold on
h2Old = plot(traj.X2(1),traj.Y2(2),'or','LineWidth',1.5);
h3Old = plot(traj.X3(1),traj.Y3(2),'og','LineWidth',1.5);
legend({'Particle 1', 'Particle 2', 'Particle
3'},'Interpreter','latex','FontSize',11);
ylim([-2 2]);
xlim([-2 2]);
xlabel('$x$','Interpreter','latex','FontSize',14);
ylabel('$y$','Interpreter','latex','FontSize',14);
title('Animation for Periodic Orbits obtained for 3-Masses under Inverse
Squared Gravity','Interpreter','latex','FontSize',14)

for i=2:n
    delete(h1Old);
    delete(h2Old);
    delete(h3Old);
    plot(traj.X1(i-1:i),traj.Y1(i-1:i),'--b','LineWidth',2);
    plot(traj.X2(i-1:i),traj.Y2(i-1:i),'--r','LineWidth',2);
    plot(traj.X3(i-1:i),traj.Y3(i-1:i),'--g','LineWidth',2);

    h1Old = plot(traj.X1(i),traj.Y1(i),'ob','LineWidth',1.5,'MarkerSize',8);
    h2Old = plot(traj.X2(i),traj.Y2(i),'or','LineWidth',1.5,'MarkerSize',8);
    h3Old = plot(traj.X3(i),traj.Y3(i),'og','LineWidth',1.5,'MarkerSize',8);
    pause(delT);
end
```
